

Simplified PRD_AI 3D Search

Acceptance Criteria

Functional Requirements

3DV Search and Select

- Provide natural language prompt in Connect's Web 3DV
- The user will be able to provide English language string inputs
- Connect's web 3DV will
 - a. The search prompt will interpret the English language search criteria and search for matching 3D model elements (e.g. Get all steel bars with cross sectional area greater than 10 sq. cm)
 - b. The model elements matching the search criteria will be selected and shown to the user in the search results
 - c. The object properties prompt will be available to the user to check. (similar to regular object selection in 3DV) .
- As a result of user writing the question and hitting enter, all the model objects that matched the search criteria will be selected on screen
 - The selection is identified as a simple and powerful first phase functionality, because the selection immediately shows that number of matching objects in the 3D viewer's context tool bar, and allows user to quickly continue the workflow by launching property panel or data table to study the details of the objects.
- Search will be only on latest version of the model
- This will be made available as an extension (Manifest URL)

Scope of the NL search

- Customer's own models that are currently displayed in the 3D screen
- The search must not find results from other models or "dream" non existing matches.

Mockups

UX inputs : [Figma mockup](#) , [Video](#)

Licensing

- Initial launch - Beta only or any other Label - to entice users and get feedback
- End state - The natural language prompt in Connect's 3DV will be enabled only for TC AI License users (new license feature to be defined)

Comparables (not substitutes or competition)

E.g. of [Kreo](#) (for estimating) : Without AI : 45 GBP ; With AI : GBP 95

OUT OF SCOPE (Functional)

(Last updated : Jul 1, 2024)

Following are out of scope -

- Project wide search of 3D model object properties
- Searching model object properties from older versions of the models
- Search based on file custom attributes , file name , file non custom attributes.
- Global search across all projects that the user is part of.
- Search based on 2D file content.
- AI 3D model object search in TCM, TCD is out of scope (at least for Phase-1 , 2)

These are good features to consider for the future (post GA) and can be considered based on customer and user feedback.

Analytics

- Track the following analytics for the natural language prompts
 - a. Number of searches (daily , weekly , monthly)
 - b. Number or search results retrieved
 - c. Duration of fetching the search results